ANLP Assignment 2025

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Use this template file for your solutions. Please answer each question within the page limits specified. For your convenience, we added half-page delimiters, please do not remove them. Different questions may be marked by different markers so your answers must be self-contained.

TEST ANSWER

1 Training your model

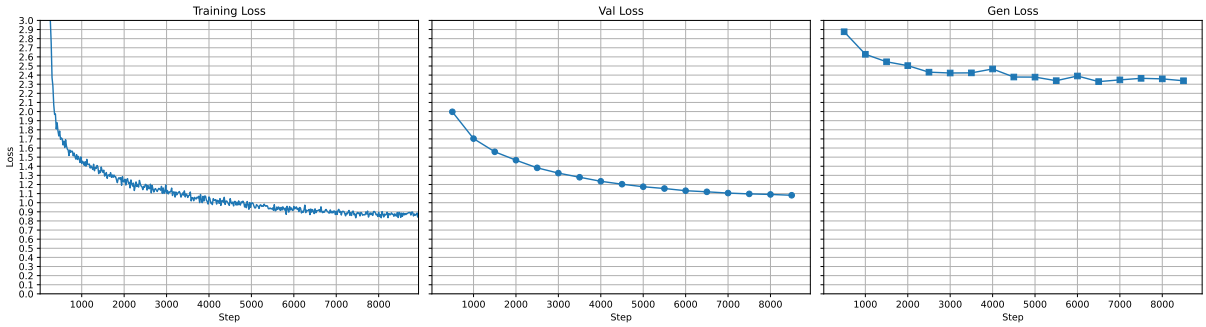


Figure 1: Plots of loss values for the training (left), evaluation (center), and generation (right) datasets.

Based on the plots of (Figure 1) we observe that the training and validation loss values decrease as the steps progress, indicating that the model learns effectively on short sentences. A well-trained model with generalization abilities would show training, validation, and generalization losses that decrease steadily and converge to similar values. However, the observed high and fluctuating generation loss on the generalization set suggests that the model does not generalize effectively, implying possible overfitting or insufficient exposure to longer sequences during training.

2 Error analysis

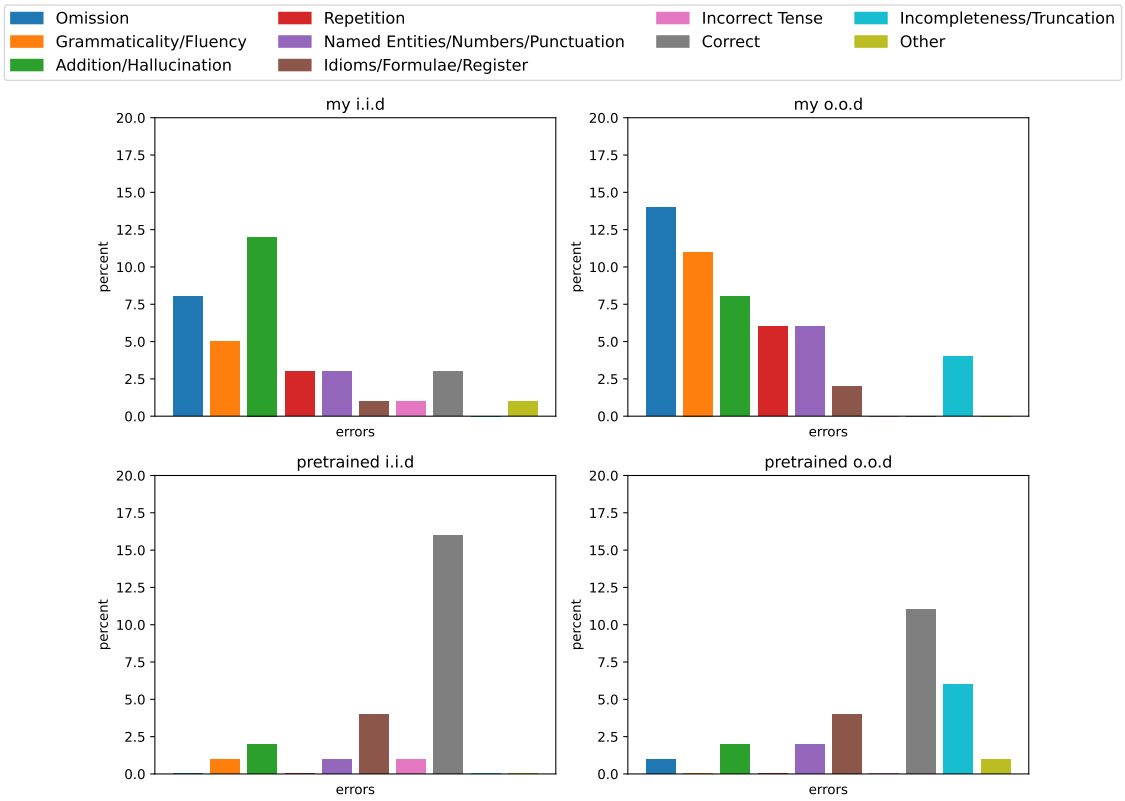


Figure 2: Distribution of translation error types across both models and datasets.

Our model on i.i.d data shows higher omission and hallucination because it has limited generalization and tends to overfit to training patterns. On the o.o.d data, unfamiliar inputs cause the model to drop information or invent content. The pretrained model on i.i.d data produces mostly correct outputs with few error occurrences, due to its exposure to larger and more varied training data, causing it to be exposed to more syntactic patterns and stabilizing generalization. Finally, on the o.o.d data, the pretrained model remains relatively accurate but displays some register, truncation, and named entity errors, as discussed in the following paragraphs.

Truncated translations occurred mostly in the pretrained model on o.o.d. examples. For the reference "The EU should not be paralyzed, though, by the opposing views adopted in the northern and southern Member States." the model stopped generating words after "... northern and". This behaviour likely stems from limited exposure to longer inputs or premature <eos> generalization. Omission was highest in our model's o.o.d split. The predicted "There are many areas of the different countries." for the reference "There are many activists in different countries who have contacted us on this matter." shows that significant content is omitted, reflecting weaker generalization to unseen syntactic structures, such as relative clauses. However, truncation and omission are mutually influencing errors, thus setting the "generation_max_length" parameter to 20 in the configuration might be responsible for the errors of both categories.

Our model exhibited frequent hallucination errors, across both splits. For example, in the o.o.d split, "One solution would be a great deal for Iraq." was predicted for the reference "A solution would be a major breakthrough in turning 2011 into a promising year for enlargement." The model inserts unsupported content, confidently generating plausible but unsupported segments and struggles to generalize to longer inputs. Fluency errors were most prevalent in our models o.o.d split. In the model's prediction "However, a re-border alternatives needs to use these work-

ers." the singular "a" is matched with a plural noun "alternatives" and the lexical verb "needs" mismatches the subject (singular/plural). Further, the model has misassigned grammatical roles and struggles handle long-distance syntactic dependencies, resulting in incoherence. The model also generated grammatically accurate translation with incorrect tense (ex. "You should do the same as we do!' instead of "...have done!"") and thus we propose that the category also includes incorrect tense errors that do not affect fluency, but meaning.

Idiomatic and register errors appeared in both models. In the pretrained i.i.d. "What are these troops gonna do?" is predicted for reference "What is this force going to do?" It produces an informal contraction. Because the model has been trained on texts of different genres, it could be choosing a higher-frequency variant, even though its register is unsuitable for a parliamentary debate. Further ,our i.i.d struggles to render a fixed idiomatic expression. The model predicted "Is it actually what is happening?" for the reference "Who are they trying to kid?" The expression is non-compositional; the meaning of this expression is not clear from the sum of its parts and the model fails to capture the idiomatic sense of 'to kid' ('to fool' or 'to deceive'). The rhetorical and skeptical tone of the idiom is lost and the model gives a neutral clarifying question. Our model fails to capture named entities and numbers. It produces "But what is the misunderstandings?" for "But where is Kofi Annan?" where the named entity is replaced with an unrelated common noun. This suggests a broader inability to maintain tokens that require exact matching.

We introduced a 'Repetition' error category due to frequent token duplication in our model's outputs across both splits. For instance, the model produced "What is the people of these people's peoples?" This is not hallucination of new content, but the repeated use of the same lexical term and it's inflected forms. This suggests that the model becomes trapped in a loop and continues to generate the same token due to incorrect correlation between words, as confirmed by the bertviz visualization of attention.

3 Evaluation metrics

Model	Split	Google BLEU	ChrF	LenR	Repr
Our Model	i.i.d	0.327	47.46	1.0270	0.0581
Our Model	o.o.d	0.157	30.46	0.5872	0.0121
Pretrained	i.i.d	0.427	62.34	1.0205	0.0503
Pretrained	o.o.d	0.379	58.54	0.9347	0.0034

Table 1: Evaluation metrics across our model and pretrained model on i.i.d and o.o.d splits.

Based on the metrics we calculated (Table 1) we observed that Google BLEU and ChrF metrics have lower values in the o.o.d split in both models, indicating poor generalization to longer sequences, while the pretrained model performs better than the custom one in both splits.

We decided to use the Google BLEU (GLEU) metric (Wu et al., 2016) which is more suitable for sentence-level translation tasks, whilst the GLEU metric was designed to be a corpus measure. Moreover, GLEU takes into account both precision and recall by calculating with respect to both the base truth and the output sequence, thus giving a more accurate score in cases of truncation and omission and, since it is symmetrical, also captures cases of addition or hallucination. GLEU also captures error of the named entities category, since it searches for exact matches and none of the above categories can be freely translated from one language to another; they follow 1 to 1 translation rules. Finally, the GLEU metric is not sensitive to permutations within n-grams (Callison-Burch et al., 2006), thus cannot capture errors in grammaticality and syntax .

BLEU will systematically under-score translations because of structural differences between the two languages. Specifically, in the case of German to English translations, one German compound word might be translated into a combination of English ones; a factor that multiplies the cases of acceptable paraphrasing, which BLEU metric penalizes by searching for exact n-gram overlap rather than semantic equivalence. However, ChrF, going on character level, captures morphological features better without being limited by word boundaries or being tokenization dependent, as well as omission and addition errors, resulting in higher scores on all models and splits compared to Google BLEU. The pretrained model produces better results according to ChrF, whilst both models perform better on shorter sentences. Although all models receive higher values in the ChrF metric, the results of this metric follow the same trends as those of the Google BLEU, as expected. Examining the role of the ChrF metric beyond the scope of this assignment, it is worth noting that it also captures errors in politeness markers, especially in languages where politeness is expressed by changes in suffixes and pronouns, like Greek.

For the implementation of the LenR metric, we decided to compare the output length to the reference length and not the input length to receive more reliable results. Considering the structural and morphological differences between languages that lead to differences in the number of tokens per sentence, a comparison between the output and the reference is a more suitable metric for translation tasks. Ideally, the metric's results should equal one, thus deviations indicate a mismatch in the sentence lengths. Both models have values below one in the o.o.d split, with our model displaying significantly worse performance, which is a sign of truncation and omission in longer sentences, a result that aligns with the previous error analysis.

The RepR metric assigns higher numbers to degenerate or redundant translation performance, thus the goal is to minimize its value. The lowest values have been assigned to both model's o.o.d split. This metric is sensitive to repetitive bigrams. Although, we found most instances of repetition in our error analysis in the o.o.d split of our model than in the i.i.d split, these are instances of unigram token repetition (e.g. "I also support cooperation cooperation and Parliament would be grateful.") and not bigram repetition, which explains the difference in the results of our error analysis and the actual metric.

References

Callison-Burch, C., Osborne, M., & Koehn, P. (2006). Re-evaluating the role of BLEU in machine translation research. *Proceedings of the 11th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 249–256.

Wu, Y., Schuster, M., Chen, Z., Le, Q. V., Norouzi, M., Macherey, W., Krikun, M., Cao, Y., Gao, Q., Macherey, K., Klingner, J., Shah, A., Johnson, M., Liu, X., Kaiser, Ł., Gouws, S., Kato, Y., Kudo, T., Kazawa, H., ... Dean, J. (2016). Google's neural machine translation system: Bridging the gap between human and machine translation.

4 Attention coverage

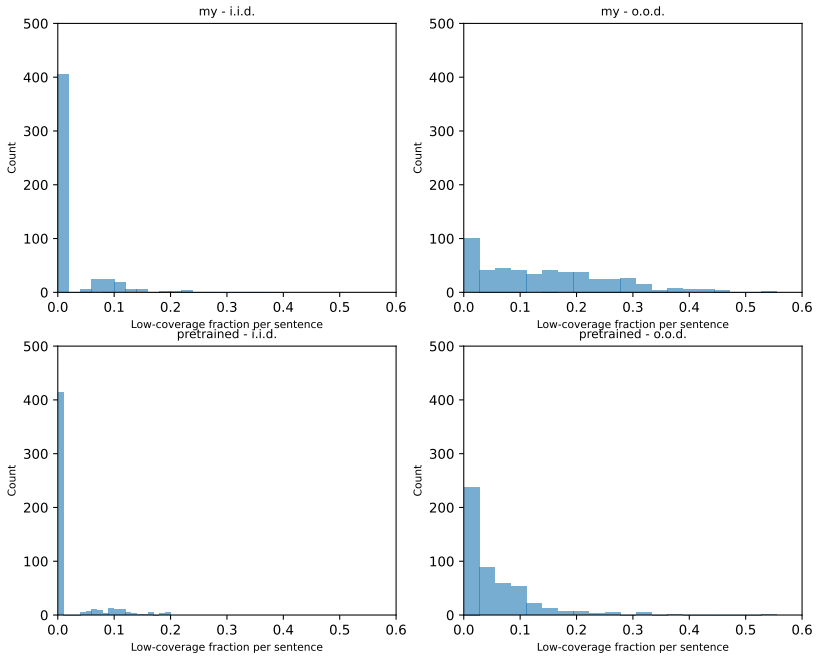


Figure 3: Plots of attention coverage per model and per dataset.

To examine whether attention coverage is related to under-translation, we compared the distributions of low-coverage fractions (Figure 3) with translation quality metrics (Google BLEU (GLEU), ChrF, Length Ratio (LenR), and bigram repetition rate (RepR)). Length Ratio (LenR) directly measures whether the output is shorter than the reference, with an ideal score of 1, and is therefore a strong indicator of under-translation (Shao et al., 2024). GLEU and ChrF reflect n-gram-level and character-level overlap respectively, so they decrease when content is omitted. RepR, defined as the proportion of repeated bigrams in the output, measures degenerative looping or redundancy rather than semantic similarity; higher values indicate more repetitive, lower-quality text.

We find that low coverage is strongly associated with omission-related errors (under-translation), such as missing nouns, dropped modifiers, or truncated sentences. Sentences with high low-coverage fractions rarely show repetition or looping errors, reflected in the RepR metric. In our dataset, low coverage aligns consistently with omission and truncation errors, while repetition errors do not correlate with coverage at all.

For our model on the i.i.d. split, the low-coverage fractions are mostly close to zero. The scores (GLEU = 0.327, ChrF = 47.46, RepR = 0.0581, LenR = 1.027) indicate that the translations are reasonably adequate in length but not necessarily in quality. The LenR score being close to one shows that the model outputs a similar number of tokens as the reference. The GLEU and ChrF scores show moderate lexical and character-level overlap, and the low RepR score indicates little bigram repetition, consistent with the high attention coverage.

In contrast, the custom model performs the worst on the o.o.d. split, showing substantially higher low-coverage fractions (some above 0.5) and far fewer sentences with full coverage. The scores (GLEU = 0.157, ChrF = 30.46, RepR = 0.0121, LenR = 0.5872) confirm this. The LenR value in particular shows that translations are much shorter than their references, and the drops in GLEU and ChrF indicate reduced lexical and character-level overlap. The very low RepR score suggests that, while the model avoids repetition, the primary issue is severe omission rather than degenerative looping. All these results are consistent with under-translation.

The pretrained model exhibits the same pattern across splits. On the i.i.d. split, the low-coverage fractions are again close to zero, and the metrics indicate good overall translation quality (GLEU = 0.427, ChrF = 62.34, RepR = 0.0503, LenR = 1.020). The LenR score close to one shows appropriate output length, and the GLEU and ChrF scores are the highest among all conditions, suggesting strong lexical correspondence with the references. The RepR score remains low, indicating minimal repeated content.

In the o.o.d. data, the pretrained model shows an increase in low-coverage fractions, with many sentences exceeding 0.3. The metric scores also decline (GLEU = 0.379, ChrF = 58.54, RepR = 0.0034, LenR = 0.935), indicating mild shortening and showing little to no repetitive degeneration.

Across all configurations, lower attention coverage consistently corresponds to lower GLEU, lower ChrF, and especially lower LenR. RepR remains low in all settings, indicating that repetition is not the main type of error associated with under-translation. Under-translation presents mainly through content omission and truncation. This alignment between attention coverage and the other three metrics provides strong evidence that attention coverage is meaningfully related to under-translation.

In general, without gold references, attention coverage is harder to interpret as an indicator of under-translation. It assumes every source token should map to something in the target, but many elements have no English equivalent. For example, Japanese case particles like は (topic marker) or を (object marker) disappear in natural English translations, so low attention on them is expected rather than an error. Thus, low coverage does not always mean content was omitted. However, for German to English, most source tokens typically have a meaningful English equivalent, so low attention coverage is much more likely to reflect true omission. In this setting, low coverage serves as a strong indicator of under-translation.

References

Shao, C., Meng, F., Zeng, J., & Zhou, J. (2024). Understanding and addressing the under-translation problem from the perspective of decoding objective. https://arxiv.org/abs/2405.18922

5 Improving your model

Beams	Max Length	Google BLEU	chrF	Lenr	Repr
4	1.0	0.32725	47.46	1.02702	0.05814
4	1.2	0.32657	**47.52**	1.03273	0.05786
10	0.8	0.32634	47.19	**1.00140**	0.06011
20	0.8	**0.32785**	47.18	0.98872	0.06330
20	1.0	0.32765	47.31	1.00883	**0.05644**

Table 2: Model evaluation on I.I.D. split. Blue color indicates previous values, while bold indicates best value per metric.

For this part of the assignment, we chose to implement option 1 due to time limitations, but also to dive deeper into the Transformers documentation, experiment with different hyperparameter combinations, and examine their impact on different datasets.

To improve the LenR metric of the i.i.d split we set Max Length to 0.8 to favor shorter sentences while increasing the beam size to 10 (max sentence length) and achieved an improvement of the score (1.027 vs 1.0014) , as seen in (Table 2). An even larger, but more costly, increase of the beam size to 15, combined with a simultaneous increase of the max length to 1.2, seems to improve the model's LenR value on the o.o.d split as well (Table 3) while also improving the GLEU and ChrF metrics. It is also worth noting that while smaller max length values (0.8) favor the i.i.d set, to improve the generalization of the model it is required to increase the value to 1.2 to motivate it to generate longer sentences and match the reference length.

Beams	Max Length	Google BLEU	chrF	Lenr	Repr
4	1.0	0.1570	30.46	0.5872	0.0121
15	1.2	**0.1577**	30.86	**0.6016**	0.0126
20	1.0	0.1570	30.63	0.5869	**0.0109**
20	1.2	0.1576	**30.92**	0.5996	0.0123

Table 3: Model evaluation on O.O.D. split. Blue color indicates previous values, while bold indicates best value per metric.

By increasing the number of Beams, we expected an overall better quality of translations since when using Beam Search the model computes each time the most probable sequence of N tokens and not only the locally optimal word. Based on that, it is expected that larger beam values will generate more coherent phrases since the model does not get stuck with a suboptimal early choice. Increasing the number of beams to the max sentence length (10) for the i.i.d split did not lead to any improvements in the Google BLEU metric. However, increasing the value to 20 did have a positive change on the metric. In sentences of maximum ten words, a beam size this large does not make sense of course, so the positive impact could be the result of the max length adjustment to 0.8. Although increasing the beam size was expected to improve generalization on the o.o.d split, it did not yield significantly better results.

To improve the bigram repetition issue in the o.o.d split, we set no_repeat_ngram_size to 2. According to the results of this experiment, the RepR metric value was significantly reduced from 0.0121 to 0.0023, indicating a reduction of repetition by 83% without any significant impact on the other metrics. A potential drawback of this change is that it might deter the model from repeating bigrams that need to be repeated within the sentence (e.g. European Union).